Optimization Problems and Solutions

Gradient Descent – Question 1

Consider:

$$f(x) = x^2 + 4x + 1$$

Starting from:

$$x_0 = 5$$

Use gradient descent with:

$$\eta = 0.1$$

Find:

- Gradient formula
- x_1, x_2, x_3
- Function value after 3 iterations

Gradient Descent – Solution 1 (Part 1)

Gradient:

$$f'(x) = 2x + 4$$

Update Rule:

$$x_{k+1} = x_k - \eta f'(x_k)$$

Given:

$$x_0 = 5, \quad \eta = 0.1$$

Iteration 1:

$$f'(5) = 14$$

$$x_1 = 5 - 0.1(14) = 3.6$$

Iteration 2:

$$f'(3.6) = 11.2$$

$$x_2 = 3.6 - 0.1(11.2) = 2.48$$

Gradient Descent – Solution 1 (Part 2)

Iteration 3:

$$f'(2.48) = 8.96$$

$$x_3 = 2.48 - 0.1(8.96) = 1.584$$

Function value:

$$f(1.584) = 1.584^2 + 4(1.584) + 1$$

$$= 2.509056 + 6.336 + 1$$

$$= 9.845056$$

Therefore:

$$f(1.584) \approx 9.84$$

Gradient Descent – Question 2

Given:

$$f(x, y) = x^2 + y^2 + xy$$

Use gradient descent with:

$$\eta = 0.2$$

Starting point:

$$(1, 2)$$

Find:

- Gradient vector
- One iteration
- Whether objective decreases

Gradient Descent – Solution 2

Gradient:

$$\nabla f(x, y) = \begin{bmatrix} 2x + y \\ 2y + x \end{bmatrix}$$

At $(1, 2)$:

$$\nabla f(1, 2) = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

Update:

$$x_1 = 1 - 0.2(4) = 0.2$$

$$y_1 = 2 - 0.2(5) = 1$$

New point:

$$(0.2, 1)$$

Function values:

$$f(1, 2) = 7$$

$$f(0.2, 1) = 1.24$$

Since:

$$1.24 < 7$$

Objective decreases.

Convex Optimization – Question 1

Show whether:

$$f(x) = 3x^2 - 2x + 7$$

is convex.
Find:

- Second derivative
- Convexity condition
- Whether strictly convex

Convex Optimization – Solution 1

First derivative:

$$f'(x) = 6x - 2$$

Second derivative:

$$f''(x) = 6$$

Since:

$$f''(x) > 0$$

for all x,
the function is strictly convex.

Convex Optimization – Question 2

Solve:

$$\min_{x,y} x^2 + y^2$$

Subject to:

$$x + y = 1$$

Find:

- Lagrangian
- Stationary point
- Optimal value

Convex Optimization – Solution 2 (Part 1)

Lagrangian:

$$L(x, y, \lambda) = x^2 + y^2 + \lambda(x + y - 1)$$

Stationarity:

$$\frac{\partial L}{\partial x} = 2x + \lambda = 0$$

$$\frac{\partial L}{\partial y} = 2y + \lambda = 0$$

Therefore:

$$x = y$$

Convex Optimization – Solution 2 (Part 2)

Constraint:

$$x + y = 1$$

Substitute $x = y$:

$$x + x = 1$$

$$2x = 1$$

$$x = \frac{1}{2}$$

Hence:

$$y = \frac{1}{2}$$

$$f\left(\frac{1}{2}, \frac{1}{2}\right) = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2$$

$$= \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Minimum value:

$$\boxed{\frac{1}{2}}$$

Primal Problems – Question 1

Solve:

$$\min_x x^2 + 6x + 8$$

Subject to:

$$x \geq 0$$

Find:

- Unconstrained minimum
- Feasible solution
- Optimal value

Primal Problems – Solution 1

Derivative:

$$2x + 6 = 0$$

$$x = -3$$

But:

$$x = -3$$

is infeasible.
Boundary point:

$$x = 0$$

Function value:

$$f(0) = 8$$

Optimal solution:

$$x^* = 0$$

Minimum value:

$$8$$

Primal Problems – Question 2

Solve:

$$\max 3x + 2y$$

Subject to:

$$x + y \leq 4$$

$$x \geq 0, \quad y \geq 0$$

Use graphical method.

Primal Problems – Solution 2

Corner points:

$$(0,0), \quad (4,0), \quad (0,4)$$

Objective values:

$$f(0,0) = 0$$

$$f(4,0) = 12$$

$$f(0,4) = 8$$

Maximum occurs at:

$$(4,0)$$

Optimal value:

$$12$$

Dual Problems – Question 1

Given primal:

$$\min 2x_1 + x_2$$

Subject to:

$$x_1 + x_2 \geq 3$$

$$x_1, x_2 \geq 0$$

Formulate the dual problem.

Dual Problems – Solution 1

Dual variable:

$$y \geq 0$$

Dual:

$$\max 3y$$

Subject to:

$$y \leq 2$$

$$y \leq 1$$

Thus:

$$y^* = 1$$

Optimal value:

$$3$$

Dual Problems – Question 2

Construct dual of:

$$\max 5x_1 + 4x_2$$

Subject to:

$$6x_1 + 4x_2 \leq 24$$

$$x_1 + 2x_2 \leq 6$$

$$x_1, x_2 \geq 0$$

Dual Problems – Solution 2

Dual:

$$\min 24y_1 + 6y_2$$

Subject to:

$$6y_1 + y_2 \geq 5$$

$$4y_1 + 2y_2 \geq 4$$

$$y_1, y_2 \geq 0$$

Optimal solution:

$$x_1 = 3, \quad x_2 = \frac{3}{2}$$

Optimal value:

$$21$$

KKT Conditions – Question 1

Solve using KKT:

$$\min x^2$$

Subject to:

$$x \geq 1$$

KKT Conditions – Solution 1

Constraint:

$$1 - x \leq 0$$

Lagrangian:

$$L = x^2 + \lambda(1 - x)$$

Stationarity:

$$2x - \lambda = 0$$

Complementary Slackness:

$$\lambda(1 - x) = 0$$

Optimal point:

$$x^* = 1$$

Multiplier:

$$\lambda = 2$$

KKT Conditions – Question 2

Solve using KKT:

$$\min x^2 + y^2$$

Subject to:

$$x + y \geq 2$$

KKT Conditions – Solution 2 (Part 1)

Constraint:

$$2 - x - y \leq 0$$

Lagrangian:

$$L = x^2 + y^2 + \lambda(2 - x - y)$$

Stationarity:

$$2x - \lambda = 0$$

$$2y - \lambda = 0$$

Therefore:

$$x = y$$

KKT Conditions – Solution 2 (Part 2)

Constraint:

$$x + y = 2$$

Substitute:

$$x + x = 2$$

$$2x = 2$$

$$x = 1$$

Hence:

$$y = 1$$

Multiplier:

$$\lambda = 2$$

Optimal value:

$$f(1,1) = 1^2 + 1^2 = 2$$

Minimum value:

$$\boxed{2}$$